INCH-POUND

MIL-E-63511 (AR)
AMENDMENT 9
25 MAY 1990
SUPERSEDING
AMENDMENT 8
13 June 1989

MILITARY SPECIFICATION

ELECTRONICS-MCD ASSEMBLY FOR MINE, ANTI-TANK: BLU 91/B

This Amendment forms a part of Military Specification MIL-E-63511 (AR), dated 15 June 1982, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Preamble: Delete "Armament Research and Development Command" and substitute "Armament, Munitions and Chemical Command."

PAGE 2

SPECIFICATIONS, MILITARY:

MIL-S-19500/127: Delete in its entirety and add the following:

MIL-S-19500/444 - Semiconductor Device, Diode, Silicon, Switching Types 2N5711 and TX 2N5711"

PAGE 3

2.1.2, Other Government documents, drawings, and publications: US Army Armament Research and Development Command (ARRADCOM): Delete "and Development Command (ARRADCOM)" and substitute ", Development and Engineering Center (ARDEC)."

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PAGE 4

3.4.1 Transportation vibration. Delete in its entirety and substitute the following:

"3.4.1 Transportation vibration. The Electronics-MCD Assembly shall be safe to handle and transport. Test mines shall show no evidence of damage that will affect the function following transportation vibration test No. 119, Procedure 2 of MIL-STD-331."

PAGE 5

3.6.1 Quiescent current. Delete "... 6.05 ± 0.15 volts at ambient temperature ..." and substitute "... 6.50 ± 0.15 volts."

PAGE 9

* Add new paragraph 3.17

"3.17 Surveillance Samples. This specification contains provisions for surveillance samples. Requirements for the submission of surveillance samples by the contractor shall be as specified in the contract and paragraph 4.4.8 herein."

PAGE 10

* 4.3.2.4 Function tests. Delete "...shall have borerider clips installed and shall be..." and substitute "...shall have borerider clips, insulator discs, and borerider pins installed and shall be ...".

PAGE 14

TABLE I:

Delete "Band Printed Wiring (Dwg. 9292230)" and substitute "Wiring, Printed Flexible (Dwg. 9292230)".

PAGE 17

4.4.2.1, Resistor, Fixed Composition, 5%, 1/8W:

TITLE: Delete "1/8W".

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DRAWING NUMBER: Delete "MIL-R-39008/4" and substitute "MIL-R-39008/1/".

Part Number, Part Value: Delete in its entirety and substitute:

"Part Number 1/	Part Value
RCR05G 30 5 JR	3 M OHM
RCR05G 10 4 JR	100 K OHM
RCR05G 47 2 JR	4.7 K OHM
RCR05G 10 2 JR	1 K OHM
RCR07G 13 1 JR	130 OHM"

Major 101: Delete "MIL-R-39008/4" and substitute "MIL-R-39008/1/".

Add the following note:

"1/ Use MIL-R-39008/1 or /4 as applicable, See Dwg. Note 13".

PAGE 18

4.4.2.2, Resistor, Fixed Film:

DRAWING NUMBER: Delete "P/N 9287099-1611" and substitute "P/N 9287099-1621".

PAGE 20

4.4.2.4, Delete in its entirety and substitute with new page 20 included

PAGE 21

4.4.2.5 Semi-conductor Device, Diode, Type JANTX IN 645-1:

Drawing Number: Delete "9287168-2" and substitute "MIL-S-19500/240"

Critical 1: Delete in its entirety.

Critical 2: Delete in its entirety.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Transistor (JANTX 2N2907A and JANTX 2N2222A)				MIL-S-19500/255 or 291
					NEXT HIGHER ASSEMBLY 9330377
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 1/					
101	Bond wire not aluminum	32	32-0-1	3.2	Visual
102	Compliance with MIL-S-19500/255 or 291		--	3.2	Cert. 2/
103	Solderability (3 leads) ea.)	10	10-0-1	3.9.1	MIL-S-45743
Minor 2/					
201	Poor workmanship		65%	3.16	Visual

NOTES:
1/ Each Transistor Type.
2/ Visual evidence of JANTX marking.

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Add the following Major Categories:

"104, Missing or incorrect color band on cathode end of diode - 100% 3.2 Electrical 1/"

"105, Reverse current - 100% 3.2 MIL-STD-750, Method 4016, dc"

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PAGE 22

4.4.2.6, Diode, Zener, JANTX IN 4463 and JANTX IN 4471:

Critical 1: Delete in its entirety

Critical 2: Delete in its entirety

Critical 3: Delete in its entirety

Add the following Major Categories:

"104, Missing or incorrect color band on cathode end of diode - 100% 3.2 Electrical, 1/"

105, Reverse current - 100% 3.2 MIL-STD-750, Method 4016, dc"

106, Zener breakdown voltage V_Z - 100% 3.2 MIL-STD-750, Method 4022"

PAGE 23

4.4.2.7, Diode, Hot Carrier (Jan 1N 5711, MIL-S-19500/127): Change title to read "4.4.2.7, Diode, Hot Carrier (JAN 1N 5711, MIL-5-19500/444)"

PAGE 25

4.4.2.9, Capacitor, Fixed, Electrolytic Tantalum, Solid (82 uF):

Major 104: Delete "10-1-2" and substitute "10-2-3"

Major 106: Delete "8-0-1" and substitute "8-1-2"

PAGE 26

4.4.2.10, Capacitor, Fixed, Electrolytic Tantalum, Solid:

Major 104: Delete "10-1-2" and substitute "10-2-3".

Major 106: Delete "8-0-1" and substitute "8-1-2"

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PAGE 32

4.4.2.16, Board, Printed Wiring:

Major 102: Delete "A" and substitute "B".

Major 105: Delete "Large circular cutout, 2 places" and substitute "Diameter of circular cutout, 2 places".

Major 106: Delete "0.40%" and substitute "1/"

Add the following Note:

"1/ Printed Wiring Board sample plan per MIL-STD-105, AQL 0.40%, Inspection Level S4. From each board, randomly select and inspect 32 hole diameters. A board shall be classed defective if any hole is defective".

PAGE 34

4.4.2.1.8, Microcircuit, Digital, CMOS:

Major 101: Delete "1/" and substitute "5-1-2"

Note 1/: Delete in its entirety.

PAGE 35

* 4.4.2.19, Core, Toriod. Delete in it entirety.

PAGE 36

* 4.4.2.20, Magnetometer Assembly. Delete Major 104, 105, 106, 107 and 108 in their entirety.

PAGE 40

4.4.2.24, Receiver Coil Assembly:

Major 104: Delete "Spacing of coil... 0.40% 3.2 Gage" and substitute "Bonding of coil to flexable circuit board incorrect 0.40% 3.2 Visual".

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PAGE 42

4.4.2.26, Board, Printed Wiring:

TITLE: Delete "Board, Printed Wiring" and substitute "Wiring, Printed Flexible".

Major 103: Delete "Diameter" and substitute "Length and Width".

PAGE 45

4.4.2.29, Lid Electronics Assembly:

DRAWING NUMBER: Delete "9292243" and substitute "9292243/9378308".

Minor 203: Delete in its entirety.

PAGE 46

4.4.2.30, Housing, Electronics Assembly:

DRAWING NUMBER: Delete "9332372" and substitute "9332372/9378309".

PAGE 50

4.4.2.33, Electronic (Performance characteristics) (Ambient): Add the following Major category:

"112, TTB Osillator period, under min. 100% - 3.6.3 - 4.5.5.3"

PAGE 52

4.4.2.33, Electronic Assembly (Performance characteristics) (-40° - 5°F): Add the following Major category:

"112, TTB Oscillator period, under min. 0.40% - 3.6.3 - 4.5.5.3"

PAGE 54

4.4.2.33, Electronic Assembly (Performance characteristics) (140° + 5°F): Add the following:

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"Major 109: Delete "100%" and substitute "0.40%".

"112, TTB Oscillator period, under min. 0.40% - 3.6.3 - 4.5.5.3"

PAGE 56

4.4.2.35, Electronics-MCD Assy. (Prior to Assy of Test Mines):

Title: Delete "(Prior to Assy of Test Mines)"

Add the following Major Category:

"103 Improper soldering MCD flex cable - 0.40% 3.16 Visual"

PAGE 57

Add the following Major defect: "103 Impact failure 0-1 3.5 4.5.4"

PAGE 58

4.4.2.36, Electronics MCD Assembly:

Add the following Lot Sample Plan:

After paragraph 4.4.2.36, add paragraph 4.4.2.37, page 58a and 58b included consisting of Classification of Defects and Tests for Power Supply Molding, drawing number 9292221.

After paragraph 4.4.2.37, add paragraph 4.4.2.38, page 58c included consisting of Classification of Defects and Test for Power Supply Assembly, drawing number 9292241.

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QUALITY CONFORMANCE INSPECTION

Paragraph 4.4.2.36 Sheet 2 of 2 Dwg. 9330379

Inspection Lot Size	Sample Size	Prime Sensor	SD
501-1200	36	9	27
1201-3200	60	15	45
3201-10000	84	21	63
*10001-35000	120	30	90

TEMPERATURE CONDITIONING													ACCEPT-REJECT CRITERIA 2/			
	-40-5°F				Ambient				145+5°F							
Sample Size	Sensor	SD 1	SD 2	SD 3	Sensor	SD 1	SD 2	SD 3	Sensor	SD 1	SD 2	SD 3	Failure 1/ Dud	Early SD	Hazardous Dud	Premature Det.
36	3	3	3	3	3	3	3	3	3	3	3	3	1-2	2-3	0-1	0-1
60	5	5	5	5	5	5	5	5	5	5	5	5	2-3	4-5	0-1	0-1
84	7	7	7	7	7	7	7	7	7	7	7	7	3-4	6-7	0-1	0-1
*120	10	10	10	10	10	10	10	10	10	10	10	10	5-6	7-8	0-1	0-1

1/ Includes sensor and SD failures.
2/ See 6.9 for definitions of defects.
3/ See 4.5.4.4 for data to be recorded.
4/ See 4.5.4.1 for sensor function.
5/ See 4.5.4.2 for SD function.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)
AMENDMENT 9

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.37	Power Supply Molding				9292221
					NEXT HIGHER ASSEMBLY 9292241
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	I.D. of small center hole		0.40%	3.2	Gage
102	O.D. of bottom post (max)		0.40%	3.2	Gage
103	True position of O.D. of bottom post		0.40%	3.2	Gage
104	Width of bridge		0.40%	3.2	Gage
105	True position of bridge width		0.40%	3.2	Gage
106	I.D. of largest center hole		0.40%	3.2	Gage
107	True position of largest hole		0.40%	3.2	Gage
108	Height from bottom to top of bridge		0.40%	3.2	Gage
109	Material		0.40%	3.2	Certificate
110	Distance from center line to edge of shorting bar cavity (Section B-B), 2 places		0.40%	3.2	Gage
111	Depth of largest hole		0.40%	3.2	Gage
112	Height from bottom to top of upper post		0.40%	3.2	Gage
113	O.D. of upper post		0.40%	3.2	Gage
114	True position of upper post		0.40%	3.2	Gage
Minor					
201	Length from bridge to closed end of battery cell cavity (2 places)		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.37	Power Supply Molding	2 OF 2	9292221
			NEXT HIGHER ASSEMBLY 9292241

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
202	True position of inside width of cavity (CD203), 2 places		0.65%	3.2	Gage
203	Width of battery cavity (2 places)		0.65%	3.2	Gage
204	Width of shorting bar cavity (Sect. B-B 2 places		0.65%	3.2	Gage
205	Length of shorting bar slot, max 2 places		0.65%	3.2	Gage
206	Depth of shorting bar cavity (Sect. B-B) 2 places		0.65%	3.2	Gage
207	Width of shorting bar slot, 4 places		0.65%	3.2	Gage
208	Distance from center line to edge of shorting bar cavity		0.65%	3.2	Gage
209	Poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63511 (AR)
AMENDMENT 9

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.38	Power Supply Assembly	1 OF 1	9292241
			NEXT HIGHER ASSEMBLY: 9330377

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pull test (weld integrity)		100%	3.2	4.5.9
Major					
101	Battery cell(s) not bonded to mold		0.40%	3.2	Visual
102	Length of lead(s) extending below mold (min)		0.40%	3.2	Gage
103	Solderability, 4 cell leads	15	0-1	3.9.1	MIL-S-45743
104	Battery leaks	1/	-	3.2	4.5.9
105	Height of cell(s) in nold over max.		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.5 Test mine assembly. Change "... mine but containing inert filler, booster, and clearing charges will be used ..." to "... mine but containing inert main and booster charges, but containing no clearing charges will be used ..."

4.4.6 Launch Simulator

a. Delete "150 + 50" and substitute: "145 + 25"

* Add the following new paragraphs:

* "4.4.8 Surveillance samples. The following subassemblies or components shall be submitted to the Government as surveillance samples.

* 4.4.8.1 Electronic assembly (dwg. 9330378). The contractor shall submit forty (40) electronic assemblies, randomly selected from each months production, to the Government for surveillance testing. Each electronic assembly in the surveillance sample shall be serialized and identified by lot number. The samples shall be packaged in accordance with paragraph 5 and delivered to the Government as specified in paragraph 6.12.

* 4.4.8.1.1 Electronic assembly surveillance sample data. The following variables data, for each electronic assembly in the surveillance sample, shall be delivered to the Government as specified in paragraph 6.13:

* Paragraph 4.4.2.33, Criticals 1 through 9 inclusive (Hot, Cold, and Ambient).

* Paragraph 4.4.2.33, Majors 101, 102, and 104 through 110 inclusive (Hot, Cold, and Ambient).

The variables data shall be identified by serial number and lot number to provide traceability. This data shall be the actual data and not averages, means, standard deviations or other compilations of the data. Any data which may be of a classified nature shall be handled in accordance with the security provisions of the contract.

* 4.4.8.2 Potting material (deg. 9287148). The contractor shall submit fifty(50) tensile test samples and two(2) plaques molded per the drawing for each unique mix of potting material to the Government for surveillance

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testing. For the purposes of this specification, a unique mix of potting material shall be defined as a mix of one batch of part A from the original manufacturer with one batch of part B from the original manufacturer. The samples shall be packaged in accordance with paragraph 5.5 and delivered to the Government as specified in paragraph 6.12.

* 4.4.8.2.1 Potting material surveillance data. For each submittal of potting material, the tensile strength and hardness data corresponding to the day that the surveillance samples were prepared (see paragraph 4.4.3.1) shall be delivered to the Government in accordance with paragraph 6.13".

PAGE 61

* 4.5.4 Functional tests of electronic -MCD assemblies. Delete "... the test samples shall be launched..." and substitute "... the test samples with borerider clips, insulator discs, and borerider pins installed shall be launched...".

PAGE 63

4.5.4.3, Disposition of Duds. Delete "ARRADCOM" and substitute "to the Government"

PAGE 64

4.5.5 Electronic assembly performance tests: Delete "... of 70° + 5°F, and ..." and substitute "... of $70^{o} \pm 10^{o}$F, and ...".

PAGE 66

* 4.5.5.10, Target sensor functioning test: Delete "... for the ambient all-fire and ..." substitute "...for the all-fire and ".

PAGE 67

In Table II Delete: "1. Shorting Bar Across Firing Capacitor 6 to 9 .2 to .7 OHMS" and substitute "1. Shorting Bar Across Firing Capacitor 6 to 9 .2 to .85 OHMS."

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Delete:

"3. Fuse and ID ... .

4. ... to 9 20K ± 5% ohms" in their entirety.

and substitute:

"3. Fuze and ID check 12 to 9 20K ± 10% OHMS.

4. Fuze and ID check 11 to 9 20K ± 10% OHMS."

PAGE 68

* 5.5 Add the following paragraph:

"5.5 Packaging of potting material surveillance samples. The potting material surveillance samples shall be packaged in a Level C pack in accordance with MIL-STD-1169".

PAGE 69

Add the following new paragraphs:

"4.5.9 Power supply assembly.

4.5.9.1 Pull test of the power supply assembly leads. Each welded lead shall withstand a non-destructive pull force 5 + 2 pounds along the longitudinal axis of the lead without degradation or failure. Additionally one power supply at the start of each shift and one Power Supply every 100 manufactured shall be subjected to a destructive pull force on each of the 4 leads of 9 pounds minimum along the longitudinal axis of the lead. Evidence of residual lead material shall be present at each weld after the pull test to assure weld integrity.

"4.5.9.2 Leak test of the power supply assembly. One assembly at the start of each shift and one assembly of every 100 manufactured shall be tested for compliance with the reoss and fine leak requirements of MIL-B-48761. Any unit failing to meet the requirement shall reject the lot. (non-destructive test) (alternate lead material only).

NOTE: At the contractors option 4.5.9.1 & 4.5.9.2 may be conducted on battery and lead assemblies prior to assembly of the Power Supply Assembly dependent on the contractor process."

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5.1: Delete in its entirety and substitute the following:

"5.1 Packaging. The Electronics-MCD assembly shall be preserved and packaged in accordance with MIL-E-17555 with the following mandatory requirements:

Each individual Electronic-MCD Assembly shall be completely wrapped in metal foil (aluminum foil) to shield the units from electromagnetic and radio frequency energies.

The wrapped units shall be packaged in a corrugated fiberboard tray or box fitted with partitions and separators as required to keep the foil-wrapped units separated.

Cushioning as needed shall be used; the fiberboard trays or box shall be sealed in a barrier bag conforming to Type I, or Class F, Style 1, Specification MIL-B-117. (Material used to fabricate the barrier bag must include metal foil).

The packaging materials shall not contain silicones, sulphur compounds or polysulphides."

6.3 Submission of inspection equipment designs for approval. Delete: "to: Commander, U.S. Army...New Jersey 07801" and substitute "to the Technical Agency specified in the contract."

PAGE 69

6.4 Submission of plans for approval. Delete "Commander: ARRADCOM, ... DRDAR-QAT-M" and substitute "the Technical Agency specified in the contract".

6.5 Submission of process control documents. Delete "to: Commander, ARRADCOM, ... DRDAR-QAT-M" and substitute "to the Technical Agency specified in the contract"

6.6 Distribution of ammunition data cards, inspection and test results, process controls. Delete "Commander: ARRADCOM, ... DRDAR-QAT-M" and substitute "the Technical Agency specified in the contract"

PAGE 70

6.9 Definitions:

h. Transportation Vibration failure: Delete in its entirety and substitute the following.

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"h. Transportation Vibration failure is defined as a test mine assembly that does not meet the requirements of 3.4.1, or in the event the borerider has functioned as a result of vibration the test shall be stopped and the Government Technical Agency specified in the Contract shall be notified for disposition of the test sample."

i. Impact failure: Delete in its entirety and substitute the following:

"i. Impact failure.

(1) Critical is defined as an Electronics-MCD Assembly that sustains damage due to impact resulting in critical failure modes for the MCD Assembly moves with respect to the Electronics Assembly (Dud failure analysis only).

(2) Major is defined as an Electronics-MCD Assembly that sustains damage due to impact resulting in a dud."

6.11 Drawings. Delete in its entirety and substitute:

"6.11 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC."

PAGE 71

Add the following paragraphs:

* "6.12 Surveillance samples. Surveillance samples specified in this specification shall be delivered to the Technical Agency specified in the contract.

6.13 Submission of surveillance sample data. The surveillance sample variables data (see paragraphs 4.4.8.1.1 and 4.4.8.2.1) shall be delivered to the Technical Agency specified in the contract."

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